С§1 №5

$(x-C)^2+y^2=1$

$\frac{\partial F}{\partial C}=2(x-C)$

$\frac{\partial F}{\partial y}=2y$

$$\begin{cases} y^2+(x-C)^2-1=0 \\ 2yy'+2(x-C)=0 \end{cases}$$

$(x-C)=-yy' \Rightarrow y^2+y^2y'^2=1$

Ответ: $y^2(1+y'^2)=1$

С§2 №4

$y'\cos x+y(1+y)\sin x=0$

$\cos x\,dy+y(1+y)\sin x\,dx=0$

$\frac{dy}{y(1+y)}=-\frac{\sin x}{\cos x}dx$ $\quad y=0;\ y=-1$ – решения

$\ln\left|\frac{y}{1+y}\right|=\ln|\cos x|+C$

$\left|\frac{y}{1+y}\right|=D|\cos x| \quad D>0$

$\frac{y}{1+y}=D\cos x \quad D\neq 0$

$y(1-D\cos x)=D\cos x$

$$\begin{cases} y=\frac{D\cos x}{1-D\cos x}, & D\in\mathbb{R} \\ y=-1 \end{cases}$$

Ответ: $y=\frac{D\cos x}{1-D\cos x}$; $y=-1$

С§2 №9

$(x^2-1)y\,dx=x(x^2+1)\,dy$

$\frac{x^2-1}{x(x^2+1)}dx=\frac{dy}{y}$ $\quad y=0\ \ x=0$ – решения

$\left(\frac{2x}{x^2+1}-\frac{1}{x}\right)dx=\frac{dy}{y}$

$\ln|x^2+1|-\ln|x|=\ln|y|+C$

$\ln|y|=\ln\left|\frac{x^2+1}{x}\right|+C$

$$\begin{cases} y=D\left(x+\frac{1}{x}\right), & D\in\mathbb{R} \\ x=0 \end{cases}$$

Ответ: $y=D\left(x+\frac{1}{x}\right)$; $x=0$

С§2 №15

$$x^2(x^2+4)y' = \cos^2 y$$

$$x^2(x^2+4)\,dy = \cos^2 y\,dx$$

$$\frac{dy}{\cos^2 y} = \frac{dx}{x^2(x^2+4)} \qquad y = \frac{\pi}{2}+\pi k \text{ - решение}$$

$$\frac{dy}{\cos^2 y} = \frac{dx}{4x^2} - \frac{dx}{4(x^2+4)}$$

$$\operatorname{tg} y = -\frac{1}{4}\frac{1}{x} - \frac{\operatorname{arctg}\frac{x}{2}}{8} + C$$

Ответ: $\operatorname{tg} y = -\frac{1}{4x} - \frac{\operatorname{arctg}\frac{x}{2}}{8} + C$; $y = \frac{\pi}{2}+\pi k$

С§2 №36

$$y' + 3y^2 = 3y \qquad y(0) = \frac{1}{2}$$

$$dy = 3(y-y^2)\,dx \qquad y=0 \quad y=1 \text{ - решения}$$

$$\frac{dy}{y(1-y)} = 3\,dx$$

$$\ln|y| - \ln|y-1| = 3x + C$$

$$\left|\frac{y}{y-1}\right| = De^{3x}$$

$$\left[\begin{array}{l} \frac{y}{y-1} = De^{3x} \\ y = 1 \end{array}\right.$$

$y(0)$: $D = \frac{\frac{1}{2}}{\frac{1}{2}-1} = -1$

$$\frac{y}{y-1} = -1\cdot e^{3x}$$

$$y = -1e^{3x}(y-1)$$

$$y(1+e^{3x}) = e^{3x}$$

$$y = \frac{1}{1+e^{-3x}}$$

Ответ: $y = \frac{1}{1+e^{-3x}}$

С§2 №43

$$y = C\sin x - 2$$

$$\Phi(x;y;C) = 0 = C\sin x - 2 - y \qquad \frac{\partial\Phi}{\partial x} = C\cos x \qquad \frac{\partial\Phi}{\partial y} = -1$$

$$\begin{cases} C\cos x - y' = 0 \\ y - C\sin x + 2 \end{cases} \qquad C = \frac{y+2}{\sin x}$$

$$(y+2)\operatorname{ctg} x - y' = 0 \qquad y' \to -\frac{1}{y'} \qquad (y+2)\operatorname{ctg} x + \frac{1}{y'} = 0$$

$$y' = \frac{-\operatorname{tg} x}{(y+2)} \qquad (y+2)\,dy = -\operatorname{tg} x\,dx$$

$$y^2/2 + 2y = \ln|\cos x| + C$$

$$\cos x = De^{y^2/2+2y}$$

Ответ: $De^{y^2/2+2y}$

С§2 №72

$$(x+y+1)dx + (x-y+3)dy = 0$$

$$\begin{cases} x+y+1=0 \\ x-y+3=0 \end{cases} \quad \begin{cases} x_0 = -2 \\ y_0 = 1 \end{cases}$$

$u = x+2$

$v = y-1$

$$(u+v)du + (u-v)dv = 0$$

$v = uz$

$dv = z\,du + u\,dz$

$$(u+uz)du + (u-uz)(z\,du + u\,dz) = 0$$

$$(u+uz+uz-uz^2)du + (u^2-u^2z)dz = 0$$

$$u(1+2z-z^2)du + u^2(1-z)dz = 0 \qquad u=0$$

$$(1+2z-z^2)du + u(1-z)dz = 0$$

$$\frac{du}{u} = \frac{(z-1)dz}{1+2z-z^2} \qquad z = 1 \pm \sqrt{2}$$

$$\ln|u| = -\frac{1}{2}\int \frac{d(z^2-2z-1)}{z^2-2z-1} = -\frac{1}{2}\ln|z^2-2z-1| + C$$

$$\ln u^2 = -\ln|z^2-2z-1| + D$$

$$u^2 = \frac{F}{z^2-2z-1} \qquad F \neq 0$$

$$u^2(z^2-2z-1) = F \quad F \in \mathbb{R} \quad \begin{pmatrix} u=0 \\ z=1\pm\sqrt{2} \end{pmatrix}$$

$$z = \frac{v}{u}$$

$$u^2\left(\left(\frac{v}{u}\right)^2 - 2\left(\frac{v}{u}\right) - 1\right) = F$$

$$v^2 - 2vu - u^2 = F \qquad F \in \mathbb{R} \quad u = 0$$

$$(y-1)^2 - 2(y-1)(x+2) - (x+2)^2 = F \qquad F \in \mathbb{R};\ x = -2$$

Ответ: $(y-1)^2 - 2(y-1)(x+2) - (x+2)^2 = F;\ x = -2$

Ф №62

$y' = \cos(y-x)$

$z = y - x$

$z' = y' - 1$

$z' + 1 = \cos z$

$\frac{dz}{dx} = \cos z - 1$

$\frac{dz}{\cos z - 1} = dx$

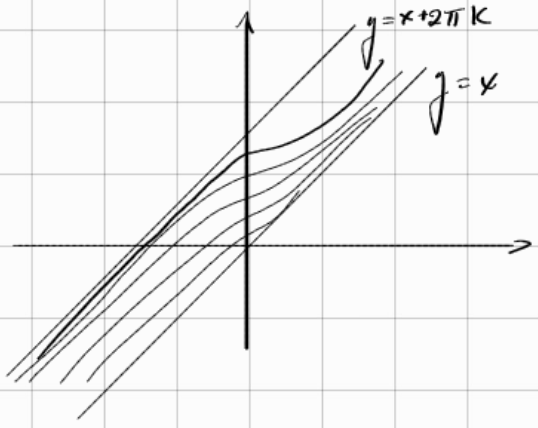
$-\int \frac{dz}{2\sin^2 \frac{z}{2}} = \int dx$

$ctg \frac{z}{2} = x + C$

$ctg \frac{y-x}{2} = x + C$

$\frac{y-x}{2} = arctg(x+C) + \pi k$

$y = 2arctg(x+C) + 2\pi k + x$



С §3 №21

$y' + 2xy = 2(1+2x^2)$

1) $y' + 2xy = 0$

$\frac{dy}{dx} = -2xy$ $\quad y = 0$ - решение

$\frac{dy}{y} = -2x\,dx$

$\ln|y| = -x^2 + C$

$y = De^{-x^2}$

2) $y' = D'e^{-x^2} + De^{-x^2}(-2x)$

$D'e^{-x^2} + \cancel{De^{-x^2}(2x)} + \cancel{2xDe^{-x^2}} = 2(1+2x^2)$

$D'e^{-x^2} = 2(1+2x^2)$

$D' = 2e^{x^2}(1+2x^2) = 2e^{x^2} + 4x^2e^{x^2}$

$D = 2e^{x^2} \cdot x + C$

$y = 2x + e^{-x^2} \cdot C$

С §3 №28 $xy' + (1+x^2)y + x = 0$

$y' + \left(\frac{1+x^2}{x}\right)y = -1$

1) $y' + \left(\frac{1+x^2}{x}\right)y = 0$

$\frac{dy}{dx} = -\frac{1+x^2}{x}y$

$\frac{dy}{y} = -\frac{1+x^2}{x}dx$

$\frac{dy}{y} = -\frac{1}{x}dx - x\,dx$ $\quad \ln|y| = -\ln|x| - \frac{x^2}{2} + C$

$y = \frac{D}{x}e^{-\frac{x^2}{2}}$

2) $y' = D'\frac{e^{-\frac{x^2}{2}}}{x} + D\frac{e^{-\frac{x^2}{2}}\cdot(-x)}{x} - D\frac{e^{-\frac{x^2}{2}}}{x^2}$

$D'\frac{e^{-\frac{x^2}{2}}}{x} + D\frac{e^{-\frac{x^2}{2}}\cdot(-x)}{x} - D\frac{e^{-\frac{x^2}{2}}}{x^2} + \left(\frac{1+x^2}{x}\right)\frac{D}{x}e^{-\frac{x^2}{2}} = -1$

$D'\frac{e^{-\frac{x^2}{2}}}{x} = -1$

$D' = -xe^{\frac{x^2}{2}}$

$D = -e^{\frac{x^2}{2}} + C$ $\quad y = -\frac{1}{x} + \frac{C}{x}e^{-\frac{x^2}{2}}$

С §3 №35

$(1+y^2)dx + (2xy-1)dy = 0$

$\begin{pmatrix} F(x;y) = C \\ F'_x dx + F'_y dy = 0 \end{pmatrix}$

$F = x + xy^2 - y = C$

С §3 №56

$xy' = 2y - 4x^2y^2$

$y' - \frac{2}{x}y = -4xy^2$ $\quad y = 0$ - решение

$z = y^{-1}$ $\quad z' = -y^{-2}y'$

$y'y^{-2} - \frac{2}{x}y^{-1} = -4x$

$-z' - \frac{2}{x} z = -4x$

$z' + \frac{2}{x} z = 4x$

1) $z' + \frac{2}{x} z = 0$

$\frac{dz}{z} = -\frac{2}{x} dx$

$\ln|z| = -2 \ln|x| + C$

$z = D \cdot \frac{1}{x^2} \quad D \in \mathbb{R}$

$z' = D' \frac{1}{x^2} - 2D \frac{1}{x^3}$

$D' \frac{1}{x^2} - 2D \frac{1}{x^3} + \frac{2}{x} D \frac{1}{x} = 4x$

$D' = 4x^3$

$D = x^4 + F$

$z = \frac{F}{x^2} + x^2 \rightarrow \frac{1}{y} = \frac{F}{x^2} + x^2$

$y = 0$

C §3 №60

$y' + y \frac{\sin x}{\cos x} + 4y^2 \sin x = 0 \quad y = 0$ - решение

$z = y^{-1} \quad z' = -\frac{y'}{y^2}$

$\frac{y'}{y^2} + \frac{\sin x}{\cos x} \cdot \frac{1}{y} = -4 \sin x$

$-z' + \frac{\sin x}{\cos x} z = -4 \sin x$

1) $z' = \frac{\sin x}{\cos x} z$

$\frac{dz}{z} = \frac{\sin x}{\cos x} dx$

$\ln|z| = -\ln|\cos x| + C$

$z = \frac{D}{\cos x}$

$z' = D' \frac{1}{\cos x} + D \cdot \frac{\sin x}{\cos^2 x}$

2) $-D' \frac{1}{\cos x} - D \frac{\sin x}{\cos^2 x} + \frac{\sin x}{\cos x} D \cdot \frac{1}{\cos x} = -4 \sin x$

$D' = 4 \sin x \cos x$

$D = 2 \sin^2 x + F$

$\frac{1}{y} = \frac{2 \sin^2 x}{\cos x} + \frac{F}{\cos x} = \frac{2(1 - \cos^2 x)}{\cos x} + \frac{F}{\cos x} = \frac{F}{\cos x} - 2 \cos x$

$y = 0$

C §3 №90

$y' + e^{-x} y^2 + y = 3e^x$

$Ae^x + e^{-x} A^2 e^{2x} + Ae^x = 3e^x$

$A = 1$

$A^2 + 2A - 3 = 0 \quad D = 16 \quad \frac{-2 \pm 4}{2} = \begin{bmatrix} 1 \\ -3 \end{bmatrix}$

$y = e^x + z(x)$

$y' = e^x + z'$

$e^x + z' + e^{-x}(e^{2x} + z^2 + 2e^x z) + e^x + z = 3e^x$

$z' + e^{-x}z^2 + 2z + z = 0$

$z' + 3z = -e^{-x}z^2$

$q = \frac{1}{z} \quad z = 0$

$q' = -\frac{1}{z^2} z'$

$-\frac{z'}{z^2} + \frac{3}{z} = -e^{-x}$

$q' - 3q = e^{-x}$

1) $q' - 3q = 0$

$\frac{dq}{dx} = 3q \quad q = 0$ – решение

$\frac{dq}{q} = 3dx$

$\ln|q| = 3x + C$

$q = De^{3x} \quad D \in \mathbb{R}$

$q' = e^{3x}D' + 3e^{3x}D$

$e^{3x}D' + 3e^{3x}D - 3De^{3x} = e^{-x}$

$e^{3x}D' = e^{-x}$

$D' = e^{-4x} \quad D = \frac{e^{-4x}}{-4} + F$

$q = e^{3x}\left(\frac{e^{-4x}}{-4} + F\right) = e^{3x}F + \frac{e^{-x}}{-4}$

$z = \frac{1}{e^{3x}F - \frac{e^{-x}}{4}}; \quad z = 0$

$y = z + e^x$

С §4 № 11

$(y^2 + \ln x)dx + (2xy - \ln y)dy = 0$

$\frac{\partial P}{\partial y} = 2y \quad \frac{\partial Q}{\partial x} = 2y$

$u = xy^2 + x(\ln x - 1) + y(\ln y - 1) = \text{const}$

С §4 № 20

$2xy\,dx - (x^2 - 2y^3)dy = 0$

$\frac{\partial P}{\partial y} = 2x \quad \frac{\partial Q}{\partial x} = -2x$

$2xy\,x^\alpha y^\beta dx + (2y^3 - x^2)x^\alpha y^\beta dy = 0$

$2x^{\alpha+1}y^{\beta+1}dx + (2x^\alpha y^{\beta+3} - x^{\alpha+2}y^\beta)dy = 0$

$2x^{\alpha+1}(\beta+1)y^\beta = 2\alpha x^{\alpha-1}y^{\beta+3} - (\alpha+2)x^{\alpha+1}y^\beta$

$\begin{cases} \alpha = 0 \\ 2(\beta+1) = -2 \end{cases} \quad \beta = -2$

$2\frac{x}{y}dx-\left(\frac{x^2}{y^2}-2y\right)dy=0$ $y=0$ - решение

$u=\frac{x^2}{y}+y^2=const$

С§4 №25

$2(y^2-x^2y)dx+x^3dy=0$

$\left(2x^\alpha y^{\beta+2}-2x^{\alpha+2}y^{\beta+1}\right)dx+x^{\alpha+3}y^\beta dy=0$

$2(\beta+2)x^\alpha y^{\beta+1}-2(\beta+1)x^{\alpha+2}y^\beta=(\alpha+3)x^{\alpha+2}y^\beta$

$\beta=-2$ $-2(\beta+1)=(\alpha+3)$

$\alpha=-1$

$x=0$, $y=0$ - решения

$\frac{2}{xy^2}(y^2-x^2y)dx+\frac{x^3}{xy^2}dy=0$

$\left(\frac{2}{x}-\frac{2x}{y}\right)dx+\frac{x^2}{y^2}dy$

$-\frac{x^2}{y}+2\ln|x|=const$

С§4 №50

$x^2y^3dx+(x^3y^2+x)dy$

$\frac{\partial P}{\partial y}=3x^2y^2$ $\frac{\partial Q}{\partial x}=3x^2y^2+1$

$x^{\alpha+2}y^{\beta+2}dx+\left(x^{\alpha+3}y^{\beta+2}+x^{\alpha+1}y^\beta\right)dx=0$

$(\beta+3)x^{\alpha+2}y^{\beta+2}=(\alpha+3)x^{\alpha+2}y^{\beta+2}+(\alpha+1)x^\alpha y^\beta$

$\alpha=-1$

$\beta=-1$

$xy^2dx+\left(x^2y+\frac{1}{y}\right)dy$ $x=0$, $y=0$ - решение

$u=\frac{x^2y^2}{2}+\ln y=const$

Ф №150

$(1-2xy)y'=y(y-1)$

$(y^2-y)dx+(2xy-1)dy=0$ $y=0$ - решение

$y(y)(y^2-y)dx+(2xy-1)y(y)dy=0$

$y'(y)(y^2-y)+y(y)(2y-1)=y(y)2y$

$y'(y)(y^2-y)-y(y)=0$

$y'+\frac{y}{y^2-y}=0$ $y=0$ - решение

$\frac{dy}{y}=\frac{dy}{y^2-y}$ $\ln|y|=-\ln|y|+\ln|1-y|+C$

$y=D\frac{1-y}{y}$ $]D=1$

$(y^2-y)dx+(2xy-1)dy=0$ $\left|\cdot\frac{1-y}{y^2}\right.$ $y=1;0$ - решение

$(1-2y+y^2)dx+\left(\frac{1}{y}-2x-1+2xy\right)dy=0$

$$\begin{cases} u = x - 2xy + y^2 x + \ln|y| - y = const \\ y = 1 \\ y = 0 \end{cases}$$

С87 № 12

$yy'' = 2y' + 2y'^2$

$z = y' \quad y'' = zz'$

$yzz' = 2z + 2z^2 \quad y = 0$ решение

$z = 0$

$z' = \frac{2z + 2z^2}{yz}$

$z' = \frac{2}{y} + \frac{2}{y} \cdot z$

$z' - \frac{2}{y} z = \frac{2}{y}$

1) $z' = \frac{2}{y} z$

$\frac{dz}{z} = 2\frac{dy}{y}$

$z = y^2 D \quad D \in R$

2) $D'y^2 + 2yD - \frac{2}{y} Dy^2 = \frac{2}{y}$

$Dy^2 = \frac{2}{y}$

$D' = \frac{2}{y^3} \quad D = -\frac{1}{y^2} + C$

$y' = y^2\left(-\frac{1}{y^2} + C\right)$

$y' = y^2(-1 + Cy^2)$

$y' = (-1 + Cy^2)$

$\frac{dy}{Cy^2 - 1} = dx$

1. $C = 0 \quad dy = -dx \quad y + x = C_1$

2. $C > 0 \quad \frac{1}{Cy^2 - 1} = \frac{1}{2\sqrt{C}}\left(\frac{1}{y - \frac{1}{\sqrt{C}}} - \frac{1}{y + \frac{1}{\sqrt{C}}}\right)$

$\frac{1}{2\sqrt{C}} \ln\left|\frac{y - \frac{1}{\sqrt{C}}}{y + \frac{1}{\sqrt{C}}}\right| = x + C_2$

3. $C < 0$

$-\frac{1}{\sqrt{C}} \operatorname{arctg} \sqrt{C} y = x + C_3$

$y = const$

С§7 №28

$2yy'' = y'^2(3-4yy')$ $\quad y(4)=1,\ y'(4)=-1$

$z = y' \quad y'' = zz'$

$2yzz' = z^2(3-4yz^2) \quad | \ z=0$

$2yz' = z(3-4yz^2)$

$z' = \frac{3z-4yz^3}{2y} = \frac{3z}{2y} - 2z^3$

$t = \frac{1}{z^2} \quad t' = -\frac{z'}{2z^3}$

$-\frac{1}{2}t' - \frac{3}{2y}t = -2$

$t' + \frac{3}{y}t = 4$

1) $t' + \frac{3}{y}t = 0 \quad \ln|t| = -3\ln|y| + c$

$t = \frac{D}{y^3} \quad D \in \mathbb{R}$

2) $\frac{D'}{y^3} - 3\frac{D}{y^4} + \frac{3}{y}\frac{D}{y^3} = 4$

$\frac{D'}{y^3} = 4 \quad D = y^4 + C_1$

$t = \frac{y^4 + C_1}{y^3}$

$\frac{1}{z^2} = \frac{y^4 + C_1}{y^3}$

$y'(4) = -1$

$y(4) = 1 \quad 1 = \frac{1+C_1}{1} \quad C_1 = 0$

$\frac{1}{z^2} = y \quad \frac{1}{y'^2} = y \quad \frac{1}{y'} = \pm\sqrt{y} \quad y' < 0 \Rightarrow \frac{1}{y'} = -\sqrt{y}$

$-\sqrt{y}\,dy = dx$

$-\frac{2}{3}y^{\frac{3}{2}} = x + C_2$

$-\frac{2}{3} = 4 + C_2$

$C_2 = -\frac{14}{3}$

$y = -\frac{3}{2}\left(x - \frac{14}{3}\right)^{\frac{2}{3}}$

С§7 №54

$yy'' = (1-x)y'^2 \quad y(1) = -1,\ y'(1) = 2$

одно отн $y \quad y' = yz \quad y'' = y(z' + z^2)$

$yy(z' + z^2) = (1-x)y^2z^2$

$z' + z^2 = (1-x)z^2$

$z' = -xz^2$

$\frac{dz}{z^2} = -x\,dx$

$-\frac{1}{z} = -\frac{x^2}{2} + C_1$

$\frac{1}{z} = \frac{y}{y_1} = \frac{x^2}{2} + C_1$

$\frac{1}{2} = \frac{1}{2} + C_1 \quad C_1 = 0$

$\frac{y}{y_1} = \frac{x^2}{2} \qquad \frac{dy}{y} = \frac{2}{x^2}\,dx$

$\ln|y| = -\frac{2}{x} + C_2$

$C_2 = 2$

$\ln|y| = 2 - \frac{2}{x}$

С§7 №65

$x^2y'' + 2x^2yy' + 2xy^2 - 2y = 0$

а) $y(1) = 2$, $y'(1) = 0$ б) $y(1) = -1$, $y'(1) = 1$

$\frac{d}{dx}(x^2y' + x^2y^2 - 2xy) = 0$

$x^2y' + x^2y^2 - 2xy = C_1$

Заметим, что 😭😭😭

а) $C_1 = 0$

$x^2y' + x^2y^2 - 2xy = 0$

$z = xy \quad y = \frac{z}{x} \quad y' = \frac{z'}{x} - \frac{z}{x^2}$

$xz' - z + z^2 - 2z = 0$

$xz' + z^2 - 3z = 0$

$\frac{dz}{3z - z^2} = \frac{dx}{x}$

$\frac{1}{3}\ln\frac{|z|}{|3-z|} = \ln|x| + C_2$

$\frac{z}{3-z} = Dx^3$

$\frac{1\cdot 2}{3 - 1\cdot 2} = D\cdot 1 \Rightarrow D = 2$

$\frac{z}{3-z} = 2x^3$

$\frac{xy}{3 - xy} = 2x^3$

$xy = 6x^3 - 2x^4y$

$y(1 + 2x^3) = 6x^2$

б) $1 + 1 + 2 = C_1 = 4$

$x^2y' + x^2y^2 - 2xy = 4$

$xz' + z^2 - 3z = 4$

$\frac{dz}{3z - z^2 - 4} = \frac{dx}{x}$

$\frac{1}{5}\ln\left|\frac{z+1}{4-z}\right| = \ln|x| + C_3$

$\frac{z+1}{4-z} = D_2x^5$

$\frac{0}{4+1} = D_2\cdot 1 \quad D_2 = 0$

$z + 1 = 0 \quad xy + 1 = 0$

Т1 $y''e^y + (y')^2e^y = 2x$

$(y' e^y)' = 2x$

$y' e^y = x^2 + C$

$e^y dy = (x^2 + C) dx$

$e^y = \frac{x^3}{3} + Cx + D$

Ф. 225 (а; б)

а) $\begin{cases} y' = 2xy + y^2 \\ y(x_0) = y_0 \end{cases}$ $(x_0; y_0) \in \mathbb{R}^2$

б) $\begin{cases} y' = 2 + \sqrt[3]{y - 2x} \\ y(x_0) = y_0 \end{cases}$ $y_0 \neq 2x_0$

Ф 228 (а; б)

а) $\begin{cases} y'' = \operatorname{tg} y + \sqrt[3]{x} \\ y(x_0) = y_0 \\ y'(x_0) = y_0' \end{cases}$

$f = \operatorname{tg} y + \sqrt[3]{x}$ $y_0 \neq \frac{\pi}{2} + \pi k$, $x_0; y_0'$ – любое

б) $\begin{cases} (x+1) y'' = y + \sqrt{y} \\ y(x_0) = y_0 \\ y'(x_0) = y_0' \end{cases}$

$x_0 \neq -1$, $y_0 \geq 0$, y_0' – любое

Ф. 229

а) $\begin{cases} y' = x + y^2 \\ y_1(x_0) = y_0 \\ y_2(x_0) = y_0 \end{cases}$ нужен ответ, что $\exists$ 2 решения задачи Коши. Это невозможно

б) $\begin{cases} y'' = x + y^2 \\ y(x_0) = y_0 \\ y'(x_0) = y_0' \end{cases}$

Да, могут. Если $(x_0; y_0)$ равны, это не означает равенство y_0' $\Rightarrow$ разные задачи Коши

Ф 230

а) $\begin{cases} y' = x + y^2 \\ y(x_0) = y_0 \end{cases}$ Нет, т.к. для З.К. $\exists!$ реш

а) $\begin{cases} y'' = x+y^2 \\ y(x_0)=y_0 \\ y'(x_0)=y_0' \end{cases}$ Нет, т.к. для ЗК $\exists!$ (!) реш.

б) $\begin{cases} y''' = x+y^2 \\ y''(x_0)=y_0'' \\ y'(x_0)=y_0' \\ y(x_0)=y_0 \end{cases}$ Может, при разных y_0''

Ф.231

$\begin{cases} y^{(n)} = x+y^2 \\ y(0)=1 \\ y'(0)=2 \end{cases}$ $n=1$ 2 3 сколько решений?

$n=1$ $y' = y^2+x$ $2=1+0$!!! нет решений

$n=2$ одно решение

$n=3$ беск. много решений

Ф.234

$y^{(n)} = f(x; y; y'; \dots; y^{(n-1)})$

$y_1 = x$ - решение

$y_2 = \sin x$

$n=1$ f_y' - нет $\Rightarrow$ 1 решение

ЗК: $y' = f(x;y)$ $y(x_0)=y_0$

$\exists\, x_0=0 \Rightarrow y_1=0$ $y_2=0$ - противоречие: 2 решения в окр-сти

$n=2$ f_y', $f_{y'}'$ - нет $\Rightarrow$ 1 решение

ЗК: $y'' = f(x;y;y')$ $y(x_0)=y_0$ $y'(x_0)=y_0'$

$\exists\, x_0=0 \Rightarrow$ $y_1=0$, $y_1'=1$; $y_2=0$, $y_2'=1$; противоречие: 2 решения в окр-сти

$n=3$ f_y', $f_{y'}'$, $f_{y''}'$ - нет $\Rightarrow \exists$ 1 реш.

ЗК: $y''' = f(x;y;y';y'')$ $y(x_0)=y_0$ $y'(x_0)=y_0'$ $y''(x_0)=y_0''$

$\exists\, x_0=0$ $y_1=0$, $y_1'=1$, $y_1''=0$; $y_2=0$, $y_2'=1$, $y_2''=0$ противоречие: 2 реш. в окр-сти

$n=4$ f'_y, $f'_{y'}$, $f'_{y''}$, $f'_{y'''}$ — $\ker \Rightarrow$ 1 реш.

ЗК: $y^{(4)} = f(x;y;y';y'';y''')$ $y(x_0)=y_0$ $y'(x_0)=y_0'$ $y''(x_0)=y_0''$ $y'''(x_0)=y_0'''$

1) $x_0=0$ $y_1=0$ $y_1'=1$ $y_1''=0$ $y_1'''=1$

$y_2=0$ $y_2'=1$ $y_2''=0$ $y_2'''=-1$

Общ реш. или совп $\Rightarrow$ 2 реш. ЗК.

Ответ: $n \geq 4$

Т2 а) $y' = -y^2$ $y(1)=-1$

$y=0$ решение

$\frac{dy}{y^2} = -dx \Rightarrow \frac{1}{y} = x+C$ $\quad y = \frac{1}{x+C}$; $y \neq 0$

теорема о сущ и ед $\forall (x_0;y_0)$ $y(1)=-1$

$y = \frac{1}{x-2}$ одна ветвь $x \in (-\infty;2)$ непродолж реш ЗК.

б) $y' = 3\sqrt[3]{y^2}$ $y(-4)=-1$ $y(2)=1$

$y=0$ реш.

$\frac{dy}{\sqrt[3]{y^2}} = 3dx$

$y^{\frac{1}{3}} = x+C$ $\quad y=(x+C)^3$

$y=0$ - особое

в $y_0=0$ не вып усл теор о сущ и ед $y(-4)=-1$

$-1=(-4+C)^3 \Rightarrow C=3$

$y(2)=1$ $\quad 1=(2+C)^3$ $\quad C=-1$

$$y = \begin{cases} (x+3)^3, & x<-3 \\ 0, & x \in [-3;1] \\ (x-1)^3, & x>1 \end{cases}$$

Ф №278

$y'^2 - 2xy' = x^2 - 4y$

$p=y'$ $\quad F(x;y;y') = 0 = x^2 - 4y + 2xy' - y'^2$

1) Найдем π-диск. мн-во : $F(x;y;p) = x^2 - 4y + 2xp - p^2$

$\frac{\partial F}{\partial p} = 2x - 2p = 0$

$$\begin{cases} p = x \\ x^2 - 4y + 2xp - p^2 = 0 \end{cases} \Rightarrow x^2 - 4y + 2x^2 - x^2 = 0 \Rightarrow y = \frac{x^2}{2}$$ ← кандидат на особое решение

2) Решаем уравнение:

$p^2 - 2xp = x^2 - 4y$

$4y = x^2 - p^2 + 2xp$

$4dy = 4p dx = 2x dx - 2p dp + 2x dp + 2p dx$

$2x dx - 2p dp + 2x dp - 2p dx = 0$

$x dx - p dp + x dp - p dx = 0$

$(x-p)dx = -(x-p)dp$

$(x-p)(dx + dp) = 0$

$\begin{cases} x = p \\ dx = -dp \end{cases} \Leftrightarrow \begin{cases} x = p \quad (1') \\ p = -x + C \quad (2') \end{cases}$

1': $x^2 - 2x^2 = x^2 - 4y$

$y = \frac{x^2}{2}$

2': $(C-x)^2 - 2x(C-x) = x^2 - 4y$

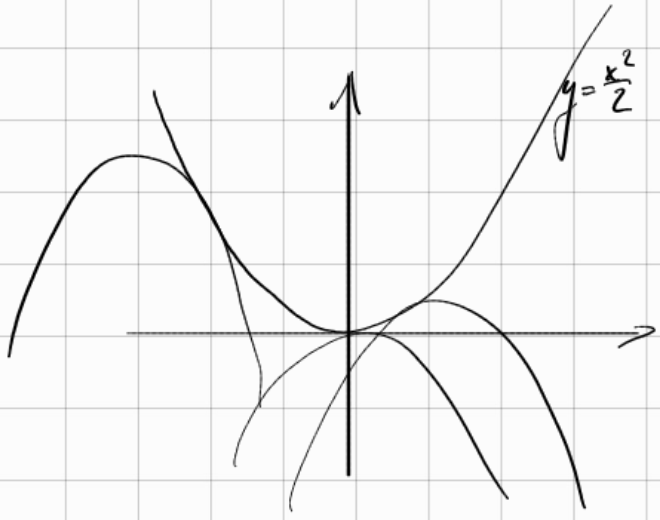
$C^2 - 2Cx + x^2 - 2xC + 2x^2 = x^2 - 4y$

$4y = C^2 - 2(C-x)^2$

$\begin{cases} -\frac{C^2}{4} + Cx - \frac{x^2}{2} = \frac{x^2}{2} \\ C - x = x \end{cases}$

$x = \frac{C}{2}$

$-\frac{C^2}{4} + \frac{C^2}{2} - \frac{C^2}{8} = \frac{C^2}{8}$ - верно - $y = \frac{x^2}{2}$ - особое



Ф. №287

$y = xy' - y'^2$

$F(x; y; y') = 0 = xy' - y'^2 - y$

1) Найдем π-дискр. мн-во: $F(x; y; p) = xp - p^2 - y$

$\frac{\partial F}{\partial x} = x - 2p = 0$

$p = \frac{x}{2}$

$\begin{cases} xp - p^2 - y = 0 \\ p = \frac{x}{2} \end{cases} \Rightarrow \frac{x^2}{2} - \frac{x^2}{4} - y = 0 \Rightarrow y = \frac{x^2}{4}$ - кандидат на особое решение

2) Решаем уравнение

$y = xy' - y'^2$

$y = xp - p^2$

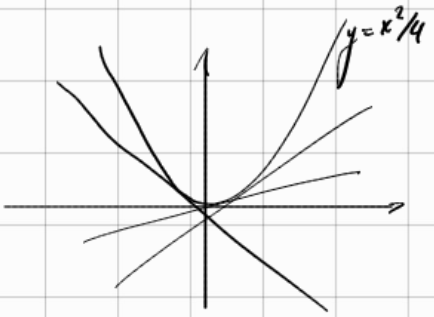
$dy = p dx = x dp + p dx - 2p dp$

$x dp - 2p dp = 0$

$(x - 2p) dp = 0$

$$\begin{cases} x-2p=0 \\ p=const \end{cases} \Leftrightarrow \begin{cases} p=\frac{x}{2} \\ p=const \end{cases}$$

$$\begin{cases} y=\frac{x^2}{4} \\ y=Cx-C^2 \end{cases}$$



3) проверим кандидата на особое решение

$$\begin{cases} \frac{x^2}{4}=Cx-C^2 \\ \frac{x}{2}=C \end{cases}$$

$\frac{x^2}{4}=\frac{x}{2}x-\frac{x^2}{4}$ – верно $\Rightarrow$ $y=\frac{x^2}{4}$ – особое

Ф. №289

$y=2xy'-4y'^3$

$F(x;y;y')=0=2xy'-4y'^3-y$

1) Найдем п-дную ур-ва: $F(x;y;p)=2xp-4p^3-y$

$$\frac{\partial F}{\partial p}=2x-12p^2=0$$

$$p^2=\frac{x}{6} \quad p=\sqrt{\frac{x}{6}}$$

$$\begin{cases} 2xp-4p^3-y=0 \\ p=\sqrt{\frac{x}{6}} \end{cases} \Rightarrow 2x\sqrt{\frac{x}{6}}-4\frac{x}{6}\sqrt{\frac{x}{6}}-y=0 \Rightarrow y=\frac{4}{3}x\sqrt{\frac{x}{6}}$$

Решаем ур-е:

$p=2p+2xp'-12p^2p'$

$p+(2x-12p^2)\frac{dp}{dx}=0$

$p\frac{dx}{dp}+2x-12p^2=0$

$\frac{dx}{dp}+\frac{2}{p}x=12p$

$p^2\frac{dx}{dp}+2px=12p^3$

$p=0$ – решение

$\frac{d}{dp}(xp^2)=12p^3$

$xp^2=3p^4+C$

$$\begin{cases} x=3p^2+\frac{C}{p^2} \\ y=2p\left(3p^2+\frac{C}{p^2}\right)-4p^3=2p^3+\frac{2C}{p} \end{cases}$$

$p=0 \Rightarrow y=0$ – особое решение

С §6 №7

$y'^3 - 3x^2y' + 4xy = 0$

1) $p = y'$

$F(x;y;p) = p^3 - 3x^2p + 4xy$

$\begin{cases} 3p^2 - 3x^2 = 0 \\ p^3 - 3x^2p + 4xy = 0 \end{cases} \Rightarrow$ $p = \pm x$

$(\pm x)^3 \mp 3x^3 + 4xy = 0$

$\mp 2x^3 + 4xy = 0 \Rightarrow y = \pm \frac{x^2}{2}$ ← кандидаты на особое решение

2) $pdx = \frac{3}{4}p\,dx + \frac{3}{4}x\,dp - \frac{3p^2}{4}dx + \frac{p^3}{4x^2}dx$

$dx\left(p - \frac{3}{4}p - \frac{p^3}{4x^2}\right) = dp\left(\frac{3}{4}x - \frac{3p^2}{4x}\right)$

$pdx\left(1 - \frac{p^2}{x^2}\right) = 3x\,dp\left(1 - \frac{p^2}{x^2}\right)$

$\left(1 - \frac{p^2}{x^2}\right)\left(pdx - 3x\,dp\right) = 0$

$\left[\begin{array}{ll} p = \pm x & (1') \\ p = C\sqrt[3]{x} & (2') \end{array}\right.$

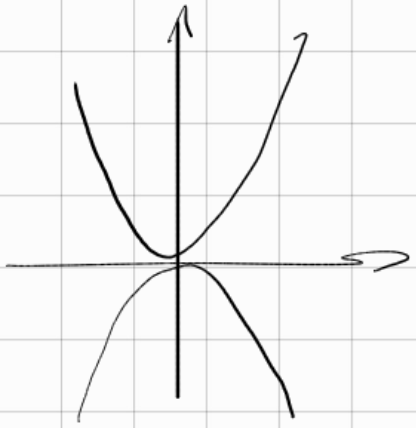
(1'): $4xy = \pm 3x^2 \mp x^3 = \pm 2x^3$

$y = \pm \frac{x^2}{2}$

(2'): $4xy = 3C\left(\sqrt[3]{x}\right)^2 - C^3x$

$y = \frac{3C|x|^{4/3} - C^3}{4}$ $C = 4d$

$y = 3d|x|^{4/3} - 16d^3$



3) Проверка

$\begin{cases} \pm \frac{x^2}{2} = 3d|x|^{4/3} - 16d^3 \\ \pm x = 4d\sqrt[3]{x} \end{cases}$

$d = \frac{\pm x}{4\sqrt[3]{x}}$

$\pm \frac{x^2}{2} = \pm \frac{3}{4}\frac{x}{\sqrt[3]{x}} \cdot x^{4/3} \mp \frac{1}{4}x^2 = \pm \frac{x^2}{2}$ ← верно! $y = \pm x^2/2$ – особое решение

С §6 №62 $(xy' + y)^2 + 20x^2y' = 0$

1) $p = y'$

$F(x;y;p) = 0 = (xp + y)^2 + 20x^2p = 0$

$\begin{cases} 2(xp+y)\cdot x + 20x^2 = 0 \\ (xp+y)^2 + 20x^2p = 0 \end{cases}$ $10x = xp + y$ $p = 10 - \frac{y}{x}$

$\left(x\left(10 - \frac{y}{x}\right) + y\right)^2 + 20x^2\left(10 - \frac{y}{x}\right) =$

$= 100x^2 + 200x^2 - 20xy = 0$

$15x = y$ – кандидат в особое решение

2) $x^2p^2 + 2xyp + y^2 + 20x^2p = 0$

$2xp^2dx + 2x^2p\,dp + 2yp\,dx + 2xp\,dy + 2xy\,dp + 2y\,dy + 40xp\,dx + 20x^2\,dp = 0$

$2xp^2dx + 2x^2p\,dp + 2yp\,dx + 2xp^2dx + 2xy\,dp + 2yp\,dx + 40xp\,dx + 20x^2\,dp = 0$

$(2xp^2 + 2yp + 2xp^2 + 2yp + 40xp)\,dx + (2x^2p + 2xy + 20x^2)\,dp = 0$

$4p(xp + y + 10x)\,dx + 2x(xp + y + 10x)\,dp = 0$

$(xp + y + 10x)(4p\,dx + 2x\,dp) = 0$

$(xy' + y)^2 + 20x^2y' = 0$

$y = \sqrt{-20x^2p} - xp$

$dy = \frac{1}{2}\sqrt{\frac{-20}{p}}x^2\,dp + \sqrt{-20p}\,x\,dx - dp - dx$

$p\,dx = \frac{1}{2}\sqrt{\frac{-20}{p}}x^2\,dp + \sqrt{-20p}\,x\,dx - dp - dx$

Т4 $\begin{cases} y = C^2 - 2(C-x)^2 \\ y = \frac{x^2}{2} \end{cases}$

а) $y(0) = -1$ $\quad -4 = -C^2 \quad C = \pm 2$

$y(5) = 6$ $\quad 24 = 4 - 2(\pm 2 - 5)^2 = 4 - 2 \cdot \begin{bmatrix} 9 \\ 49 \end{bmatrix}$ – нет решений

б) $y(0) = -1$

$y(4) = 4$ $\quad 16 = 4 - 2(\pm 2 - 4)$ – нет решений

Т5. $(y')^2 = 4y^3(1-y)$

$y' = \pm 2\sqrt{y^3(1-y)}$

$y = 1$

$y = 0$ – ещё особые

$\frac{dy}{dx} = \pm 2 y^{\frac{3}{2}}\sqrt{1-y}$

$\frac{dy}{\sqrt{y^3(1-y)}} = \pm 2\,dx$

$y = \sin^2\varphi$

$dy = 2\sin\varphi\cos\varphi = \pm 2x + C$

$2\int \frac{d\varphi}{\sin^2\varphi} = \pm 2x + C$

$-\operatorname{ctg}\varphi = \pm x + C$

$\sqrt{\frac{1}{\sin^2\varphi} - 1} = \sqrt{\frac{1-y}{y}} = \mp x + C$

$\frac{1-y}{y} = (x \pm C)^2$

$1 - y = (x+C)^2 y$

$\begin{cases} \frac{1}{(x \pm C)^2 + 1} = y \\ y = 0 \\ y = 1 \end{cases}$

проверка $0 = \frac{1}{1+(x \pm C)^2}$ неверно $y = 0$ не особое

$1 = \frac{1}{1+(x \pm C)^2}$

$0 = \frac{-2(x \pm C)}{(1+(x \pm C)^2)^2}$

$x = \mp C \quad 1 = 1$ – верно

